- **Updating Dictionary**

Python Dictionary Interview Questions & Answers

1.What is a dictionary in Python?

Answer:
A dictionary is a collection of **key-value pairs**.

- Keys are unique
- It is mutable (can be changed)

- Syntax:

dict_name = {key1: value1, key2: value2}

- **Example:**

student = {"name": "Rahul", "age": 20}

print(student)

2. How to update a value?

You can replace the value of an existing key.

- **Syntax:**

dict_name[key] = new_value

- **Example:**

student = {"age": 20}

student["age"] = 25

print(student)

3. How to add a new key?

If the key does not exist, it will be added.

- **Syntax:**

dict_name[new_key] = value

- **Example:**

```
student = {"name": "Rahul"}

student["city"] = "Ahmedabad"

print(student)
```

4. What does `.update()` do?

It updates or adds **multiple key-value pairs** at once.

- **Syntax:**

```
dict_name.update({key: value})
```

- **Example:**

```
student = {"name": "Rahul"}

student.update({"age": 30, "city": "Surat"})

print(student)
```

5. Assignment vs `.update()`

- Assignment → updates one key
- `.update()` → updates multiple keys

- Syntax:

```
d[key] = value

d.update({key1: v1, key2: v2})
```

- **Example:**

```
d = {"a": 1}

d["a"] = 5

d.update({"b": 10, "c": 15})

print(d)
```

6. Update only if key exists

Check before updating to avoid errors.

- **Syntax:**

```
if key in dict_name:
   # update
```

- Example:

```
d = {"a": 10}
if "a" in d:
   d["a"] += 5
print(d)
```

7. What is `.get()` method?

Safe way to access values. Returns default if key is missing.

- **Syntax:**

```
dict_name.get(key, default)
```

- **Example:**

```
d = {}
d["a"] = d.get("a", 0) + 10
print(d)
```

8. Update all values using loop

You can modify all values using a loop.

- **Syntax:**

```
for key in dict_name:
   dict_name[key] = value
```

- **Example:**

```
d = {"a": 1, "b": 2}

for k in d:

   d[k] += 10

print(d)
```

9. Dictionary comprehension

A short and clean way to transform a dictionary.

- **Syntax:**

```
new_dict = {k: v for k, v in dict.items()}
```

- Example:

```
d = {"a": 1, "b": 2}

new_d = {k: v + 10 for k, v in d.items()}

print(new_d)
```

10. Merge two dictionaries

Combine two dictionaries into one.

- **Syntax:**

```
d1.update(d2)
```

- **Example:**

```
d1 = {"a": 1}

d2 = {"b": 2}

d1.update(d2)

print(d1)
```